MIL-F-32G
9 February 1973
Superseding
MIL-F-32F
22 March 1965

MILITARY SPECIFICATION

FILM, PHOTOGRAPHIC, AERIAL, BLACK AND WHITE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE.-

1.1 Scope.- This specification covers the requirements for black and white roll films for aerial photography, and films for duplicating aerial photographs.

1.2 Classification.- Films covered by this specification shall be classified by alpha-numerical designators in accordance with the film classification system shown in Table I of this specification and the applicable provisions of Federal Standard 170. (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS.-

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing

FEDERAL STANDARDS

Fed Std No. 170	Film, Photographic, Black and White, Classification and Testing Methods
Fed Test Method Std No. 406	Plastics: Methods of Testing

FSC 6750

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

TABLE I

FILM CLASSIFICATION SYSTEM

SAMPLE DESIGNATOR: 1 4 3 3 2 4 • d

PRODUCT	TYPE	SUPPORT	TOTAL THICKNESS (MILS)	SPECTRAL SENSITIVITY	GAMMA	RELATIVE SENSITIVITY
1 Aerial Film	1 Negative	1 Cellulose Ester, regular	1 1.0 - 1.9	2 Blue Sensitive	2 0.51-1.00	a .57 - 1.1
	4 Duplicating	2 Cellulose Ester, Low Shrink	2 2.0 - 3.8	5 Panchromatic	3 1.01-1.50	b 1.2 - 2.1
			3 3.9 - 5.1	6 Panchromatic, Extended Red		c 2.2 - 4.2
		3 Polyester	4 5.2 - 6.5	7 Infrared	4 1.51-3.00	d 4.3 - 8.5
			5 6.6 - 8.2			e 8.6 - 17
						f 18 - 35
						g 36 - 71
						h 72 - 140
						i 141 - 280
						j 281 - 570
						k 571 - 1100
						l 1101 - 2300
						m 2301 - 4500

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

MILITARY SPECIFICATIONS

MIL-L-10547	Liners, Case Waterproof

MILITARY STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 X 48 inch Pallets
AN 7008	Spool, Aircraft Camera Film
MS26561	Can, Photographic Film Storage, for Packing Aerial Film, Outline of
MS26565	Spool, Photographic Film, Aerial
MS27575	Container, Plastic, Photographic Aerial Film, Outline of

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

GOVERMENTAL

Civil Aeronautics Board Publications

Department of Transporatation Regulations

Federal Aviation Agency Regulations

Federal Hazardous Substances Act, Public Law 86-613

Interstate Commerce Commission Regulations

Maritime Administration Regulations

Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft (DSAM 4145.3, AFM 71-4, TM 38-250, NAVAIR 15-03, and MCO P4030.19)

U.S. Postal Manual (Chapter 1, Section 137)

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Officer, Washington, D. C. 20402.)

BRITISH STANDARDS

British Standard 1380	Method for Determining the Speed of Sensitized Photographic Material; Part 1, Negative Monochrome Material for use in Daylight, dated 1962

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

AMERICAN NATIONAL STANDARDS

ANSI PH1.10	Roll Film and Leaders and Trailers for Aerial Photography
ANSI PH1.20	70mm Unperforated and Perforated Film for Cameras Other Than Motion Picture Cameras
ANSI PH1.25	Safety Photographic Film
ANSI PH1.29	Curl of Photographic Film
ANSI PH1.31	Brittleness of Photographic Film
ANSI PH1.32	Dimensional Change Characteristics of Photographic Films and Papers
ANSI PH2.33	Resolving Power of Photographic Materials
ANSI PH2.34	Speeds of Monochrome Photographic Negative Films for Aerial Photography

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS.-

3.1 Aerial Roll Film.- Aerial roll film shall consist of a darkroom loading supply of photographic film wound on spools for use in aerial cameras and on film duplicating equipment.

3.2 Spools.- Negative and duplicating films shall be furnished on Military Standard Aerial Film Spools as specified in Tables II and III, respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the current version before use.

3.3 Dimensions.-

3.3.1 Thickness.- The nominal thickness of the films furnished under each thickness designator specified herein shall fall within the limits specified in Table I. The thickness of the film furnished under each product designator shall be within plus or minus 0.5 mil of the nominal thickness of the product.

3.3.2 Width.- The width of films immediately after slitting by the manufacturer shall fall within the tolerances specified in Tables II and III. When films are furnished with opaque leader and trailer, the width of the leader and trailer shall conform to the dimensions specified in ANSI PH1.10.

3.3.3 Length.- Unless otherwise specified (See 6.2), the minimum lengths of film, leader and trailer of each respective nominal thickness to be furnished on each spool shall be not less than those specified in Tables II and III for negative and duplicating films, respectively.

3.3.3.1 Splices.- At the option of the manufacturer, film lengths greater than 1000 feet may have one internal splice. The total thickness of the splice shall not exceed the thickness of the film by more than 0.006 inch. Tensile strength of splices shall be not less than that specified in ANSI PH1.10.

3.3.4 Film Ends.- The leading end of each roll of film shall be cut in accordance with the applicable provisions of ANSI PH1.10.

3.3.5 Perforations.-

3.3.5.1 70 Millimeter Film.- Unless otherwise specified (See 6.2), all 70 millimeter films shall be furnished with Type II perforations in accordance with ANSI PH1.20.

3.3.5.2 5 Inch Film.- When perforated 5 inch films are specified in the contract or purchase order (See 6.2), perforations shall be Type II of ANSI PH1.20 except for the following dimensions:

Dimensions	Inches	Millimeters
A	4.96 ± 0.010	125.984 ± 0.254
F	4.582 ± 0.010	116.383 ± 0.254
G	Not over 0.010	Not over 0.254

3.3.6 Edge Marking.- Aerial negative roll films shall be sequentially numbered along one edge of the film within 0.075 inch of the film edge. Seventy millimeter films shall be numbered at intervals no greater than four inches. All other widths shall be numbered at intervals no greater than twelve inches. Manufacturer's commercial edge marking will be permitted provided such marking falls within the specified limit and does not interfere with the edge numbering.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

TABLE II
NEGATIVE FILM DIMENSIONS

NOMINAL WIDTH	SLITTING WIDTH (Inches)	SPOOL	OPAQUE LEADER & TRAILER	TOTAL INTEGRAL LEADER & TRAILER (FEET)	FILM LENGTH (FEET) NOMINAL 6.0 MIL THICK	FILM LENGTH (FEET) NOMINAL 4.5 MIL THICK	FILM LENGTH (FEET) NOMINAL 3.0 MIL THICK
70mm	2.754+ 0.002−	MS26565-1		3	15	20	
		MS26565-1A*		5	25		
		MS26565-2		5	50	60	100
		MS26565-3		10	100	150	200
		MS26565-4		20	250	350	500
		MS26565-4				250	
		MS26565-5		25	500	700	1,000
		MS26565-6		30	1,000	1,400	2,000
5 inch	4.960+ 0.010−	MS26565-6A*		5	25		
		MS26565-7		5	50	50	100
		MS26565-8		10	100	150	200
		MS26565-9		20	250	350	500
		MS26565-10		20	480	700	1,000
		MS26565-10				500	
		MS26565-11		20	730	850	1,500
		MS26565-12		30	970	1,400	2,000
5-1/4 inch	5.212 ±0.010	AN7008-1	4ft - 2ft		20		
7 inch	±0.010	MS26565-13	6ft-6ft		62	90	125
		MS26565-13	6ft-6ft		125	175	250
9-1/2 inch	9.460 ±0.010	MS26565-13A*		5	25		
		MS26565-14A*		5	50		
		MS26565-15A*		10	75	125	200
		MS26565-16		10	100		
		MS26565-17	10ft-6ft		150	250	400
		MS26565-17		20	180	250	400
		MS26565-18		20	230	350	500
		MS26565-19		20	370	500	700
		MS26565-20	None	None	500	700	1,000
		MS26565-21		20	730	850	1,500
		MS26565-22		30	970	1,400	2,000

*SUBJECT TO INTERNATIONAL AGREEMENTS

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the currentversion before use.

TABLE III
DUPLICATING FILM DIMENSIONS

NOMINAL WIDTH	SLITTING WIDTH (INCHES)	SPOOL	FILM LENGTH (FEET) NOMINAL 7.5 MIL THICK	FILM LENGTH (FEET) NOMINAL 4.5 MIL THICK
70mm	2.745 ±0.002	MS26565-3	100	
		MS26565-4	200	200
		MS26565-4		250
		MS26565-4		350
		MS26265-5		250
		MS26565-5	400	400
		MS26565-5		500
		MS26565-5		700
		MS26265-6	800	1,400
3-1/4 inch	3.245 ±0.005	Modified * Core **	400	100**
5 inch	4.960 ±0.010	MS26565-9	200	200
		MS26565-9		350
		MS26565-10	400	
		MS26565-10***		500
		MS26565-10		700
		MS26565-12		500
		MS26565-12		1,000
9-1/2 inch	9.460 ±0.10	MS26565-17	150	150
		MS26565-17		225
		MS26565-17		250
		MS26565-19		250
		MS26565-19	300	300
		MS26565-19		500
		MS26565-20	400	400
		MS26565-20		500
		MS26565-20		700
		MS26565-22	800	1,400

* MS26565-10 spool modified to fit the 3-1/4 inch films.
** Nominal 6.0 mil film on paperboard core with nominal 1-1/8 inch inside diameter.
*** MS26565-10 spool modified with flange diameter of 6.625 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

3.4 Materials.- All materials used in manufacture of the film, containers, and packaging shall be entirely suitable for the intended purpose and shall be free from chemical or radio active contamination that might fog or otherwise damage the film.

3.5 Film.-The film shall consist essentially of a thin layer of light sensitive emulsion of gelatin and silver halide coated onto a transparent supporting base.

3.6 Film Base.- The film base shall be a clear smooth sheet of cellulose ester or synthetic resin safety film of uniform thickness and optical transmittance.

3.6.1 Safety.- Safety film shall comply with the requirements of ANSI PH1.25.

3.6.2 Antihalation.- Film base materials shall be permanently tinted with a neutral grey dye or coated with suitable antihalation protection to inhibit the scatter of light within the film. Except for products with a permanently dyed film base, dyes or other materials used for antihalation protection shall dissolve or bleach out during conventional processing of the film without deleterious effect on the film or the processing solutions.

3.6.3 Base Density.-The base density of fixed out samples of unexposed aerial films coated on clear base and dyed base material shall not exceed 0.10 and 0.30, respectively.

3.7 Physical Characteristics.-

3.7.1 Tensile Strength.-The tensile strength of films coated on support types 1 and 3 shall be not less than 10,000 pounds per square inch in the lengthwise direction. Films coated on support type 2 shall have a tensile strength of not less than 9,000 pounds per square inch.

3.7.2 Curl.- The curl of unprocessed films at 70F and 50 percent relative humidity shall not exceed a value of 100 as specified in ANSI PH1.29. The curl of processed films shall not exceed a value of 60 at 70F and 50 percent relative humidity after processing and conditioning as specified herein.

3.7.3 Brittleness.-The wedge brittleness value, determined in accordance with Method B of ANSI PH1.31, shall be not greater than 0.445 inch for films with total thickness designators 1 through 4 (See Table I), nor greater than 0.65 inch for films with a thickness designator of 5.

3.7.4 Dimensional Stability.- Dimensional change characteristics in any direction shall not exceed the values specified in Table IV for each support type and total thickness designator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

TABLE IV
DIMENSIONAL CHARACTERISTICS (MAXIMUM PERMISSIBLE CHANGES)

Film Classification System Designator for Support Type Total Thickness	Humidity Coefficient of Expansion % of %RH	Differential Humidity Coefficient of Expansion % of %RH	Thermal Coefficient of Linear Expansion % of % ^{o}F	Dimensional Change Due to Processing %	Differential Dimensional Change Due to Processing %	Dimensional Change Due to Processing & Aging %	Differential Dimensional Change Due to Processing & Aging %
XX-14-XXX	0.010	N/A	0.004	0.15	N/A	0.35	N/A
XX-24-XXX	0.010	0.001	0.005	0.100	0.03	0.20	0.05
XX-32-XXX	0.004	0.001	0.0015	0.05	0.02	0.10	0.03
XX-33-XXX	0.003	0.001	0.0015	0.04	0.02	0.08	0.03
XX-35-XXX	0.002	0.001	0.0015	0.02	0.02	0.05	0.03

NOTE: The numerical digits (shows as XX's) have not been included in the Table. A complete designator may be selected from Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

3.7.5 Bonding.- The emulsion shall be firmly bonded to the film base in a manner to withstand processing and prolonged storage without peeling or separating from the base.

3.7.6 Physical Defects.- The film shall be free from scratches, pin holes, or other defects that would impair the ability of the film to record a photographic image or create a change in density across a uniform density area.

3.8 Emulsion.-

3.8.1 Coating.-The emulsion shall be evenly coated in a manner to ensure uniform density distribution over the entire film area.

3.8.2 Sensitometric Characteristics.-Sensitometric characteristics of the film shall conform to Table V when exposed and processed in accordance with the sensitometric test conditions specified in Table VIII.

3.8.2.1 Gamma.- The slope of the straight line portion of the characteristic curve of the film furnished under each sensitometric designator shall fall within the respective gamma range specified in Table V.

3.8.2.2 Relative Sensitivity.-The relative sensitivity of the film furnished under each sensitometric designator shall fall within the respective limits specified in Table V.

3.8.2.3 Exposure Indexes and Film Speeds.- Exposure indexes and film speeds for negative aerial roll films as given in Table VI and defined herein shall be legibly marked on film package labels (See Figure 1), to guide the user in setting exposure meters, calculators, and exposure control systems.

3.8.2.3.1 Aerial Film Speed.- The aerial film speed for negative films intended for use in aerial cameras shall be defined as the American National Standard Aerial Film Speed, determined in accordance with the standard processing method specified in ANSI PH2.34.

3.8.2.3.2 Effective Aerial Film Speed.-Emulsions for which the aerial film speed cannot be determined in accordance with the American National Standard processing method shall be rated in "Effective Aerial Film Speeds" as defined in ANSI PH2.34, using the sensitometric test conditions specified in Table VIII.

3.8.2.3.3 Aerial Exposure Index.- The Aerial Exposure Index for negative films intended for use in aerial cameras shall be determined as the reciprocal of twice the exposure in meter-candle seconds at the point on the toe of the characteristic curve where the slope is 0.6 gamma when exposed and processed as specified in Table VIII.

3.8.2.3.4 British Exposure Index.- The British Exposure Index shall be determined in accordance with the applicable provisions of British Standard 1380.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

TABLE V
SENSITOMETRIC CHARACTERISTICS OF AERIAL NEGATIVE FILM AND DUPLICATING FILM

SENSITOMETRIC DESIGNATOR	RELATIVE SENSITIVITY RANGE METHOD E*	GAMMA	SPECTRAL SENSITIVITY (MAXIMUM) WRATTEN FILTER FACTORS					MAXIMUM FOG	MAXIMUM INCUBATION FOG	D-MAX (MINIMUM)	MINIMUM RESOLVING POWER (1/mm)
			25	47	58	70	89B				
64g	50	2.30-2.90	5	10	12	16	----	.05	.25	3.50	180
64i	200	1.80-2.85	6	10	16	25	----	.10	.25	3.20	90
64k	800	1.50-2.10	5	10	16	25	----	.20	.40	2.90	63
	METHOD F (Infrared)										
74f	18-35	1.51-3.00	—	—	—	—	3	.15	.35	2.30	70
	METHOD B (Positive)										
24a	.57-1.1	1.40-1.75						.05	.10	2.75	285
23d	4.3-8.5	0.90-1.40						.05	.10	2.75	140

*Plus or minus 1/2 stop.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

TABLE VI

EXPOSURE INDEXES AND FILM SPEEDS

Applicable Designators Relative Sensitivity	EXPOSURE INDEXES AND FILM SPEEDS			
	(AEI) Aerial Exposure Index	(EAFS) Effective Aerial Film Speed	(AFS) Aerial Film Speed	(BEI) British Exposure Index
g	20	—	50	24°
h	64	—	100	29°
i	80	—	200	30°
j	100	—	400	31°
k	200	—	800	34°
(Infrared)				
f	100*	200*	—	—

* Calculated without a filter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

3.8.2.4 Filter Factor.- Filter factors for filters, applicable to film furnished under each spectral sensitivity designator, as defined in Table I, shall not exceed those specified in Table V for each respective sensitometric designator.

3.8.2.5 Fog.- Fog as defined in Federal Standard 170 shall not exceed the density specified in Table V for each respective sensitometric designator.

3.8.2.6 Incubation Fog.- Incubation fog, as tested in accordance with FED STD 170, shall not exceed the maximum fog levels indicated in Table V for the product.

3.8.2.7 Maximum Density.-The maximum density attainable on the film furnished under each sensitometric designator shall be not less than the density specified in Table V when exposed and processed in accordance with the sensitometric test conditions specified herein.

3.8.2.8 Resolving Power.- The resolving power of the film shall be defined as the American National Standard High Contrast Resolving Power in accordance with ANSI PH2.33. The resolving power of the film for each sensitometric designator shall be not less than the value specified in Table V when the film is processed in accordance with the respective method under which each film is listed in the table.

3.8.3 Machine Processing.-Aerial roll film shall withstand processing in a roller transport machine such as the Processing Machine, Photographic, EH-38 at machine speeds from 5 to 20 feet per minute using commercially available high contrast chemistry appropriate to the film used with solution temperature of 90F.

3.8.3.1 Drying.- The processed film shall dry to equilibrium with an atmosphere of 58 percent, plus or minus 2 percent, relative humidity at a temperature of not less than 70F and not more than 80F after drying at a rate of 20 feet per minute with the dryer operating in the environment specified herein.

3.9 Workmanship.- Workmanship shall be in accordance with good commercial practice for the manufacture of gelatin and silver halide photographic film.

4. QUALITY ASSURANCE PROVISIONS.-

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

4.2 Sampling for Inspection and Acceptance.- Unless otherwise specified, sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105. The manufacturer's sampling and inspection procedures may be utilized providing such sampling and inspection methods will assure quality equal to or better than that obtained by the following provisions set forth in MIL-STD-105, and provided the manufacturer's methods have been approved by the procuring activity. For purposes of sampling, an inspection lot shall consist of all materials of the same emulsion number and of the same film classification designator submitted for inspection and delivery at one time.

4.3 Inspection of Materials and Components.- In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.1 Examination of the End Item.- Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. Not more than three rolls of film shall be drawn from any one shipping container for each lot, for each designator type and emulsion number of film offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL-STD-105 shall be expressed in units of rolls for examinations under 4.3.2, 4.3.3 and 4.3.4 and in units of shipping containers for examination in 4.3.5.

4.3.2 Examination of the End Item for Defects in Appearance and Workmanship.- The sample unit for this examination shall be nine linear inches of roll film. Not more than three samples shall be drawn from any roll.

4.3.3 Examination of the End Item for Dimensional Defects.- The unit for the examination shall be two and one-half linear feet of film from each sample as applicable.

EXAMINE	DEFECT
Roll Film Width	Not within width tolerance provided in Tables II and III.
Perforations (When required)	Not perforated as required by specification. Perforations not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

4.3.4 Examination of the End Item for Defects in Construction and Length.- The sample unit for this examination shall be one roll of film.

EXAMINE	DEFECT
Roll	Film blocks. Not wound on specified core or spool; dimensions of core or spool not as specified, core or spool cracked, broken, chipped, mutilated or collapsed. Does not contain specified leader and trailer. Length of leader and trailer not as specified; leader and trailer ends not cut in accordance with ANSI PH1.10

4.3.5 Examination of Preparation for Delivery.- An examination will be made to determine that marking, materials, workmanship, contents and weights comply with the requirements of Section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

EXAMINE	DEFECT
Packaging and Preservation	Not as specified, not in accordance with contract requirements. Unit packages not uniform. Container of improper construction.
Packing	Not packed as specified. Packing material not as specified; closures not accomplished by specified or required methods or materials. Number of rolls per container less than indicated.
Markings (Interior and Exterior)	Interior labels not correct color. Data on labels not in accordance with contract specifications. Labels not properly attached. Exterior labels or markings illegible, incorrect, incomplete, or not in accordance with contract requirements.
Weight (Interior and Exterior)	Gross or net weight exceeds contract requirements.

4.3.6 Inspection Levels and Acceptable Quality Levels (AQLs) for Examination.- The inspection levels, for determining the sample size and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

Examination	Inspection Level	AQL
4.3.2	II	1.0
4.3.3	S-3	1.0
4.3.4	S-2	4.0
4.3.5	S-2	4.0

4.3.7 Sampling for Tests.- The sample unit shall consist of a sufficient length of film to conduct the tests. Sample size shall be selected at Inspection Level S-2. Any failure of the film to meet all the sensitometric requirements with a maximum variation of plus or minus 0.025 LogE for the relative sensitivity requirements within the samples tested, and the physical requirements shall be cause for rejection of the lot.

4.4 Quality Conformance Inspection.- Test methods applicable to the requirements specified herein are listed in Table VII.

TABLE VII
QUALITY CONFORMANCE TESTS

Test	Test Method	Requirement
Sensitometric Tests	4.4.1.1	
Gamma	4.4.1.2	3.8.2.1
Relative Sensitivity	4.4.1.3	3.8.2.2
Aerial Film Speed	4.4.1.3.1	3.8.2.3.1
Infrared Film Speed	4.4.1.3.3	3.8.2.2
Duplicating Film Speed	4.4.1.3.4	3.8.2.2
Spectral Sensitivity	4.4.1.4	3.8.2.4
Fog	4.4.1.5	3.8.2.5
Incubation Fog	4.4.1.6	3.8.2.6
Maximum Density	4.4.1.7	3.8.2.7
Base Density	4.4.1.8	3.6.3
Uniformity of Coating	4.4.1.9	3.8.1
Resolving Power	4.4.1.10	3.8.2.8
Physical Tests		
Safety	4.4.2.1	3.6.1
Tensile Strength	4.4.2.2	3.7.1
Curl	4.4.2.3	3.7.2
Brittleness	4.4.2.4	3.7.3
Coefficient of Expansion	4.4.2.5	3.7.4
Processing Shrinkage	4.4.2.6	3.7.4
Aging Shrinkage	4.4.2.7	3.7.4
Physical Defects	4.4.2.8	3.6.2, 3.7.5, 3.7.6, 3.8.3
Drying	4.4.2.9	3.8.3.1
Dimensions	4.4.2.10	3.3.1, 3.3.2, 3.3.3, 3.3.3.1, 3.3.4, 3.3.5.1, 3.3.5.2, 3.3.6

4.4.1 Sensitometric Tests.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

4.4.1.1 Test Conditions.- Sensitometric test strips cut from sample films selected as specified herein shall be exposed and processed as specified in Table VIII and in accordance with the applicable provisions of Federal Standard 170.

TABLE VIII
SENSITOMETRIC TEST CONDITIONS

Sensitometric Designator	Illuminant (Fed Std 170)	Exposure Time (Seconds)	Developer (Fed Std 170)	Developing Time (Minutes)
54g	1	1/40-1/120	1	8
64i	1	1/40-1/120	1	8
74f	3	1/40-1/120	1	8
24a	2	1/2	2	4
23d	2	1/5	3	7

4.4.1.2 Gamma.-Verify that the slope of the straight line portion of the characteristic curve falls within the limits specified in Table V.

4.4.1.3 Relative Sensitivity.- Determine the relative sensitivity of the film in accordance with Method E, F, or B of Federal Standard 170, as specified in Table V, and verify that the value obtained falls within the limits specified in Table V.

4.4.1.3.1 Aerial Film Speed.-The relative sensitivity of films for which Table VIII specifies processing in developer formula Number 1 of Federal Standard 170 for 8 minutes shall be determined in accordance with the American National Standard Aerial Film Speed Method as defined in ANSI PH2.34. Verify that the aerial film speed marked on the labels of unit packages complies with the applicable requirements of the American National Standard Aerial Film Speed Scale as specified in ANSI PH2.34.

4.4.1.3.2 Effective Aerial Film Speed.- The relative sensitivity of films for which the American National Standard sensitometric processing conditions do not apply shall be determined in accordance with the "effective aerial film speed" method of ANSI PH2.34, using the sensitometric test conditions specified in Table VIII. Verify that the word "effective" precedes the aerial film speed marking on labels of unit packages to alert the user that the speed rating applies to other than the American National Standard test conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

4.4.1.3.3 Infrared Film Speed.- Using the sensitometric test conditions specified in Table VIII, determine the relative sensitivity of infrared film in accordance with Method F of Federal Standard 170. Verify that labels of unit packages are properly marked to show the film speed.

4.4.1.3.4 Duplicating Film Speed.- Using the sensitometric test conditions specified in Table VIII, determine the relative sensitivity of infrared films in accordance with Method F of Federal Standard 170. Verify that the labels of unit packages are properly marked to show the film speed.

4.4.14 Spectral Sensitivity.- Determine the filter factors in accordance with Federal Standard 170 and verify that the values obtained do not exceed those specified in Table V.

4.4.1.5 Fog.-Determine the fog in accordance with the applicable provisions of Federal Standard 170 and verify that the value obtained does not exceed the value specified in Table V.

4.4.1.6 Incubation Fog.- Determine the incubation fog in accordance with the procedure specified in Federal Standard 170 and verify that the value obtained does not exceed the fog level specified in Table V.

4.4.1.7 Maximum Density.-Verify that the maximum density attained is not less than the value specified in Table V.

4.4.1.8 Base Density.-Determine the base density in accordance with the procedure specified in Federal Standard 170 and verify that the values obtained do not exceed 0.10 for clear base film, nor 0.30 for dyed base film.

4.4.1.9 Uniformity of Coating.- Expose samples of not less than one square foot of film to a uniform source of suitable intensity to produce an overall density within the range of 0.8 to 1.1 on the processed film. Examine the processed film visually for evidence of mottle, streaks, or areas of uneven density attributable to faulty emulsion coating.

4.4.1.10 Resolving Power.- Determine the resolving power of the film in accordance with ANSI PH2.33 and verify that the values obtained are not less than those specified in Table V.

4.4.2 Physical Tests.-

4.4.2.1 Safety.- Conduct the safety tests in accordance with ANSI PH1.25 and determine that the ignition time, burning time, and nitrogen content are within the specified limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

4.4.2.2 Tensile Strength.- Conduct the test in accordance with Method 1013 of Federal Test Method Standard No. 406. Verify that the tensile strength in the lengthwise direction is not less than 10,000 pounds per square inch for films coated on support types 1 and 3, nor less than 9,000 pounds per square inch for films on support type 2.

4.4.2.3 Curl.- Using test specimens cut from unprocessed and processed samples, conduct the curl test in accordance with Method A or Method B of ANSI PH1.29. Determine that the curl does not exceed a value of 100 for unprocessed film, nor a value of 60 for processed film at a temperature of 70F and relative humidity of 50 percent.

4.4.2.4 Brittleness.- Using specimens cut from unprocessed samples, conduct the wedge brittleness test in accordance with Method B of ANSI PH1.31. Determine that the brittleness value does not exceed 0.445 inch for films with total thickness designators 1 through 4, nor 0.65 inch for films with a thickness designator of 5.

4.4.2.5 Coefficient of Expansion.- Using specimens cut from unprocessed samples, determine the humidity coefficient of expansion and thermal coefficient of expansion in accordance with the applicable procedures of ANSI PH1.32. Verify that the values obtained do not exceed those specified in Table IV.

4.4.2.6 Shrinkage Due to Processing.- Using test samples exposed to produce an overall maximum density approximating that specified in Table V, process in accordance with Table VIII. Conduct the test in accordance with Method A or B of ANSI PH1.32 and determine that the values obtained do not exceed the values specified in Table IV.

4.4.2.7 Shrinkage Due to Processing and Aging.-Using samples exposed and processed as specified herein for the processing shrinkage test, conduct the test in accordance with the applicable provisions of ANSI PH1.32 for dry accelerated aging conditions with an aging time of one week. Verify that the values obtained do not exceed those specified in Table IV.

4.4.2.8 Physical Defects.- Using test samples exposed to produce an overall density of approximately 0.80 to 1.1 when processed in a high contrast developer suitable for the film used at half the maximum machine speed recommended by the film manufacturer, or five feet per minute, whichever is greater, process the film in a roller transport processor such as the Processing Machine, Photographic EH-38 at a solution temperature of 90F, plus or minus one degree F and dryer adjusted to ensure adequate drying. Examine the processed film for evidence of pinholes, scratches, blisters, peeling, separation of emulsion from the base, antihalation dye residue, or any other physical defect that would render the film unusable for its intended purpose.

Source: https://assist.dla.mil -- Downloaded: 20260-16T04:26Z
Check the source to verify that this is the current version before use.

4.4.2.9 Drying.- Repeat the physical defects tests with the dryer operating at a temperature of 140F, plus or minus 5 degrees F in an ambient temperature of not less than 70F nor more than 80F at an ambient relative humidity of not less than 55 nor more than 60 percent and a rate of 20 feet per minute, plus or minus 2 feet per minute. Cut a specimen of about 2 by 3 inches from the dried sample as soon as possible after drying. Suspend the specimen in the atmosphere within a closed vessel in which the relative humidity has been stabilized at 58 percent at a temperature of 70F by a saturated solution of sodium bromide in the bottom of the vessel. The vessel may be a wide mouth pint jar with a tight fitting lid pierced by a 3/8 inch hole to permit weighing the specimen without removing it from the vessel. Place about 100 cc of saturated solution of sodium bromide in the vessel and close the hole in the lid with a rubber disc pierced by a steel wire with a hook on each end. Place the vessel under an analytical balance and adjust the length of the wire to permit weighing the specimen without removing it from the controlled environment by raising the wire just enough to lift the wire and rubber disc clear of the jar lid. Weigh the specimen, along with the wire and rubber disc immediately upon suspending it in the vessel, then lower the specimen so that the disc closes the hole. Allow the specimen to remain three minutes in the controlled environment and reweigh. A loss of weight indicates that the moisture content of the specimen exceeds 58 percent and fails to meet the drying requirement specified herein. Specimens that show no change or that gain weight between weighings shall be considered sufficiently dry for the purpose of this specification.

4.4.2.10 Dimensional Inspection.- After the respective slitting, perforating, edge marking, and spooling operations, measure the thickness, width, perforations, length of film, length of leader and trailer, and dimensions and lateral runout of the spools. Determine that the dimensions of the film fall within the tolerances specified in Tables II and III, that leaders and trailers comply with ANSI PH1.10, that perforations and edge markings are within the specified tolerances, and that the dimensions and lateral runout of spools and cores comply with the applicable requirements of the respective spool standards specified in Tables II and III.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and Packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2).

5.1.1 Level A Package.- Each roll of film shall be preserved in a metal can conforming to the applicable requirments of MS26561 or a durable opaque plastic container conforming to MS27575. Can and container shall be sealed

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

with Type III, Class I water resistant tape in accordance with Federal Specification PPP-T-60, except that the tape may be omitted from self-sealing containers on which the lids are properly designed to provide an effective seal against penetration of moisture. Each sealed can or plastic container shall be packaged in a corrugated paperboard carton Type CF, Class Domestic of PPP-B-636, except that cartons will not be required over unit containers of high impact plastic which have been specifically designed for safe shipment of aerial roll films.

5.1.2 Level C Package.- Each roll of film shall be preserved and packaged in accordance with the manufacturer's commercial practice with sufficient protection to assure that the product is in acceptable condition at the initial destination.

5.2 Packing.-(See 6.2)

5.2.1 Level A Pack.-Film packaged as specified in 5.1 shall be packed in weather resistant fiberboard shipping containers, Type CF, Variety SW, Grade V3c; Type SF, Variety SW, Grade V2s, V3s, W5s, or W6s in accordance with the applicable provisions of PPP-B-636 for special requirements and Type I loads, in triple wall corrugated fiberboard boxes, Class 2, Style optional in accordance with PPP-B-640, or in Class 2, Style 2, 2-1/2, 3, 4, or 4-1/4 nailed wood boxes in accordance with the applicable provisions of PPP-B-621. Containers conforming to PPP-B-621 shall be fitted with waterproof case liners in accordance with the applicable provisions of MIL-L-10547. Gross weights of shipping containers shall not exceed the limits imposed by the applicable container specifications.

5.2.2 Level C Pack.- Films packaged as specified in 5.1 shall be packed in exterior type shipping containers in accordance with the manufacturer's commercial practice. Containers shall be packed in a manner to ensure safe transportation at the lowest rate to the point of delivery and shall meet the requirements of the following rules and regulations as applicable to the modes of transporatation:

a. Civil Aeronautics Board Publications
b. Department of Transportation Regulations
c. Federal Aviation Agency Regulations
d. Federal Hazardous Substances Act, Public Law 86-613
e. Interstate Commerce Commission Regulations
f. Maritime Administration Regulations
g. Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft
h. U.S. Postal Manual

5.2.3 Refrigerated Transportation.- Unless otherwise specified (See 6.2), film shipments shall be refrigerated during transit at a temperature not to exceed 50F.

5.2.4 Palletization.-When specified by the procuring activity, exterior containers shall be palletized in accordance with MIL-STD-147, or as otherwise specified by the procuring activity. The method of bonding the palletized load shall be limited to strapping.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

5.3 Labeling.- Film containers and unit packages shall be legibly labeled to identify the characteristics of the product. Marking on the labels shall include the appropriate military type designator in accordance with the classification system specified in Federal Standard 170 and Table I of this specification, the nominal film size, manufacturer's emulsion number, the Aerial Film Speed, the Effective Aerial Film Speed, the Aerial Exposure Index, the British Exposure Index, Federal Stock Number as assigned by the procuring activity (See 6.2), and a military expiration date extending 24 months from date of manufacture, except for infrared films on which the expiration date shall be marked in accordance with the manufacturer's commercial practice. A suitable data block shall be provided on the label, with marked spaces for recording job number, subject, date, roll number, strip number, camera, focal length, altitude, exposure, and remarks. Labels shall conform to the dimensions and arrangements shown in Figure 1, except that the dimensions of labels for film sizes of 7 inches or less in width, may be smaller in size and may omit the data block.

5.3.1 Color of Labels.- Labels shall be legibly printed in bold type on colored paper stock according to the spectral sensitivity designator of the film.

Spectral Sensitivity	Label Color
2 Blue Sensitive	White
5 Panchromatic	Green
6 Panchromatic, Extended Red	Light Green
7 Infrared	Light Blue

5.4 Marking.- Shipping containers shall be marked in accordance with the applicable provisions of MIL-STD-129. Marking shall include: "Film, Photographic, Aerial, (Black and White)", the film classification designator number, nominal film size, emulsion number, Federal Stock Number, and military expiration date.

5.4.1 Precautionary Marking.- The following precautionary notice shall be prominently marked on the shipping container in bold face red letters.

PHOTO SENSITIVE MATERIALS
STORE IN COOL DRY LOCATION AWAY
FROM RADIOACTIVE MATERIALS

6. NOTES.-

6.1 Intended Use.- The films covered by this specification are intended for use in aerial photography and for duplication of aerial films.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

DARK ROOM LOADING	DATA
DESIGNATOR MANUFACTURER'S NAME MFR'S COMMERCIAL DESIGNATION	JOB NUMBER _____ SUBJECT _____ DATE _____ ROLL NUMBER _____ STRIP NUMBER _____ CAMERA _____ FOCAL LENGTH _____ ALTITUDE _____ EXPOSURE _____ REMARKS _____
AERIAL FILM SPEED _____ EFFECTIVE AERIAL FILM SPEED _____ AERIAL EXPOSURE INDEX _____ BRITISH EXPOSURE INDEX _____ FILM WIDTH _____ FILM LENGTH _____ TOTAL SELF LEADER & TRAILER _____ TOTAL LENGTH _____ STOCK NUMBER _____ FILM, PHOTOGRAPHIC, AERIAL BLACK AND WHITE SPECIFICATION NUMBER MIL-F-32G EXPIRATION DATE _____ EMULSION NUMBER _____	

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the current version before use.

6.2 Ordering Data.- Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Film classification system designator (See Table I, 3.8.2.1 through 3.8.2.7 and 6.3)
(c) Federal Stock Number (See 5.3)
(d) Perforations (See 3.3.5)
(e) Number of rolls required
(f) Nominal length, width and thickness (See Tables II and III)
(g) Level of packaging and packing (See Section 5)
(h) Requirements for refrigerated transportation (See 5.2.4)

6.3 Film Classification System Designator.- The following example illustrates the application of Table I:

Sample Designator: 11-33-64i

Interpretation:

11 designates aerial negative film.

33 designates polyester support, not less than 3.9 nor more than 5.1 mils thick.

64 designates a panchromatic emulsion with extended red sensitivity which is capable of attaining a gamma of at least 1.51, but not greater than 3.00 when processed as specified herein.

i designates a relative sensitivity of not less than 141, nor more than 280 when determined in accordance with the American National Standard Aerial Film Speed Method as specified in ANSI PH2.34.

Orders should state the complete designator with no omissions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:26Z
Check the source to verify that this is the current version before use.

6.4 International Agreements.- Certain provisions of this specification (See Table II) are the subject of international standardization agreement STANAG 3178. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

CUSTODIANS

Army - EL
Navy - AS
Air Force - 70

REVIEW ACTIVITIES

Air Force - 11

USER ACTIVITIES

Army - AV
Navy - MC

PREPARING ACTIVITY

Navy - AS

Project No. 6750-0241

Source: https://assist.dla.mil -- Downloaded: 202640-16T04:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-F-32G FILM, PHOTOGRAPHIC, AERIAL, BLACK AND WHITE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, PA 19112

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DOD 316

U.S.MAIL

Engineering Specifications and Standards Department (Code X)
Naval Air Engineering Center
Philadelphia, PA 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:26Z
Check the source to verify that this is the current version before use.